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Lung Adenocarcinoma

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Continuing Education Activity

Lung adenocarcinoma is the most common primary lung cancer seen in the United States. It falls under the umbrella of non-small cell lung cancer (NSCLC) and has a strong association with previous smoking. While incidence and mortality have declined, it remains the leading cause of cancer death in the United States. Adenocarcinoma of the lung usually evolves from the mucosal glands and represents about 40% of all lung cancers. It is the most common subtype to be diagnosed in people who have never smoked. Lung adenocarcinoma usually occurs in the lung periphery, and in many cases, may be found in scars or areas of chronic inflammation. This activity describes the pathophysiology of adenocarcinoma of the lung and highlights the role of the interprofessional team in its management.

Objectives:

- Describe the pathophysiology of adenocarcinoma of the lung.
- Review the workup of a patient with adenocarcinoma of the lung.
- Summarize the treatment options for adenocarcinoma of the lung.
- Outline the importance of improving care coordination among interprofessional team members to improve outcomes for patients affected by lung adenocarcinoma.

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Introduction

Lung adenocarcinoma is the most common primary lung cancer seen in the United States. It falls under the umbrella of non-small cell lung cancer (NSCLC) and has a strong association with previous smoking. While incidence and mortality have declined, it remains the leading cause of cancer death in the United States. Adenocarcinoma of the lung usually evolves from the mucosal glands and represents about 40% of all lung cancers. It is the most common subtype to be diagnosed in people who have never smoked. Lung adenocarcinoma usually occurs in the lung periphery, and in many cases, may be found in scars or areas of chronic inflammation.[1]

Etiology

By far, the main risk factor for any lung cancer, including adenocarcinoma, is smoking tobacco. Due to numerous carcinogens present in tobacco smoke, primary or secondary exposure increases risk proportional to the amount of exposure.

Other risk factors include a family history of lung cancer, or occupational exposure to other agents such as silica, asbestos, radon, heavy metals, and diesel fumes, though these are less

prevalent. Resultant genetic mutations in the p53 gene are the most frequent cause of tumorigenesis in NSCLC in 52% of cases.[2]

Epidemiology

Although incidence and mortality have declined since the 1980s, in 2015 there were 221,200 new cases of lung and bronchial cancers and more than 158,000 lung cancer deaths representing the most common cause of cancer death.

Lung cancer is also widespread globally. Despite new treatments, the 5-year survival is less than 12% to 15%. Over the past 4 decades, there has been a marked increased in lung adenocarcinoma in women, and this has been linked to smoking. The mean age of diagnosis of lung adenocarcinoma is 71 years, and this particular cancer is very rare before the age of 20. In the last 2 decades, adenocarcinoma has replaced squamous cell cancer of the lung as the most prevalent non-small cell cancer.[3]

Pathophysiology

Lung adenocarcinoma is classified into 4 types: adenocarcinoma in situ (AIS), minimally invasive adenocarcinoma (MIA), invasive adenocarcinoma, and variants of adenocarcinoma. Of these AIS and MIA have better outcomes when resected early. Local spread may involve spread directly to the pleura, diaphragm, pericardium, or bronchi with advanced disease spreading to the mediastinum, great vessels, trachea, esophagus, vertebral column, or adjacent lobe. Lymph node metastasis occurs in peribronchial lymph nodes before moving to mediastinal or subcarinal nodes and then the contralateral lung. Distant metastasis includes extension to a contralateral lobe, pleural nodules, malignant pleural or pericardial effusion, or any distant site such as the brain, bones, or liver. There is a subset of NSCLC that have mutations in epidermal growth factor receptor (EGFR), which sensitizes them to tyrosine kinase inhibitors, as well as anaplastic lymphoma kinase (ALK) fusion oncogene rearrangements.[4][5]

Histopathology

When the lung is biopsied, the histological analysis will reveal a tumor arising from the bronchial glands. Mucus production is also quite evident. The new World Health Organization (WHO) classification subclassifies adenocarcinomas as arising from the following:

- Acinar
- Papillary
- Bronchoalveolar
- Mucus-secreting

Except for Stage 1 lung cancer, adenocarcinoma carries a much worse prognosis than squamous cell cancer.

History and Physical

Symptoms and physical signs are dependent on the stage of lung cancer. The earliest stages are often asymptomatic, with nodules found incidentally on radiographic images testing for other disease processes. Later stage disease may present with nonspecific symptoms such as a cough, hemoptysis, or unintentional weight loss. If the patient presents with a pleural effusion, he or she

may have shortness of breath with decreased breath sounds. The vast majority of patients will have a smoking history and may have other associated diseases such as chronic obstructive pulmonary disease (COPD) or a family history of lung cancer.

A significant number of patients with lung adenocarcinoma will present with a locoregional spread that may include symptoms from:

- Superior vena cava obstruction
- Phrenic nerve palsy
- Horner syndrome
- Compression of brachial plexus
- Pericardial effusion

Paraneoplastic syndromes are rare with adenocarcinoma but may include:

- Cushing syndrome
- Eaton Lambert syndrome
- Hypercalcemia
- SIADH
- Hypertrophic osteoarthropathy

Evaluation

High-risk patients like current and former heavy smokers are recommended to undergo screening with low-dose CT scan by the US Preventative Services Task Force.[6][7]

If a lung nodule is found, the next step depends on the and imaging characteristics of the lung nodule. If a nodule is suspicious for lung cancer, PET/CT may be performed followed by biopsy or surgical excision. Based upon National Comprehensive Cancer Network Guidelines the next step is a full CT of the thorax and abdomen with contrast (including adrenals), bronchoscopy, mediastinal lymph node evaluation, complete blood count, and blood chemistry profile.

Brain MRI is recommended for those with Stage II, III, or IV disease to rule out metastasis. These results are then synthesized to generate a clinical stage to guide treatment.

If bone metastases are suspected, then a bone scan should be obtained.

PET scan is usually used to assess for recurrence of the disease.

Sputum cytology is rarely helpful as most adenocarcinomas are peripheral lesions.

Needle thoracentesis is done when an effusion is seen. It can be both diagnostic and therapeutic.

Additionally, any patients being considered for surgical resection should undergo pulmonary function testing to determine the feasibility of lung resection based on predicted postoperative lung function.

If the CT scan reveals mediastinal nodes, then a mediastinoscopy or thoracoscopy is recommended to stage the patient.

Staging of the patient is mandatory before recommending any treatment.[8][9]

Treatment / Management

Stage I/II/IIIA

These are limited invasive tumors (NO) or limited nodal disease. The tumor is assessed for resectability, and if operable, surgical resection is recommended with lymph node sampling. If the patient is not an operative candidate, then definitive radiotherapy with possible adjuvant chemotherapy may be performed if the patient has positive nodes or is high risk. Some specific invasive tumors may be treated with neoadjuvant chemoradiation before resection.[10][11][12]

Stage IIIB and Stage IV

These stages involve mediastinal, subcarinal, and/or contralateral nodes and metastatic disease. These stages are considered unresectable and are treated with chemoradiation. Some extrapulmonary sites may be treated as well for palliation.

The pathologic specimen is tested for EGFR sensitizing mutations and ALK mutation. Those that are positive for EGFR may be treated with tyrosine kinase inhibitors, while those exhibiting the ALK mutation may be treated with ALK inhibitors as first-line chemotherapy. If the tumor is EGFR and ALK-negative, first-line chemotherapy is usually a platinum-based doublet, with bevacizumab as a possible third agent.

After treatment, patients need surveillance with CT Chest every six to 12 months for two years and annual low-dose CT. This should be done more frequently in those with residual disease. Locoregional occurrence may be treatable. Options include external beam radiation therapy, resection, chemotherapy, and photodynamic therapy depending on where the lesion has recurred and the associated symptoms.[13]

Differential Diagnosis

- Benign lung lesions
- Granulomas
- Hamartoma
- Metastatic lesion
- Pneumonia

Surgical Oncology

Surgery is the treatment for patients with stage I to stage IIIA adenocarcinoma of the lung. Lobectomy or a pneumonectomy are often performed.

Since these patients have a high risk of relapse, adjuvant chemotherapy is now standard.

Radiation Oncology

Radiation therapy is only an option for patients who are not surgical candidates.

Medical Oncology

Because the majority of lung adenocarcinomas are incurable or advanced, chemotherapy is often used. Despite significant advances in chemotherapy, the survival of most patients with lung adenocarcinoma remains abysmal. Platinum-based regimens remain the mainstay of chemotherapy.

For patients with metastatic disease, molecular targeted therapy is being offered, but the results are not spectacular. At best the survival is increased by a few months, but the medications can cost over $20,000 a month.

Staging

Staging of Lung Adenocarcinoma

Occult cancer: TX N0 M0

Primary cancer not found. No lymph node or distant metastasis.

Stage 0

- T is N 0 M 0

Stage I

IA1

- T1mi N 0 M 0
- T1a N 0 M 0

IA2

- T1b N 0 M 0

IA3

- T1c N 0 M 0

IB

- T2a N 0 M 0

Stage II

IIA

- T2b N 0 M 0

IIB

- T1a/T1b/T1c N 1 M 0
- T2a/T2b N 1 M 0
- T3 N 0 M 0

Stage III

IIIA

- T1a/T2b/T2c N 2 M 0
- T2a/T2b N2 M 0
- T3 N 1 M 0
- T4 N 0/N 1 M 0

IIIB

- T1a/T1b/T1c N 3 M 0
- T2a/T2b N 3 M0
- T3 N 2 M 0
- T4 N 2 M 0

IIIC

- T3 N 3 M 0
- T4 N 3 M 0

Stage IV

IVA

- Any T/Any N M1a or M1b

IVB

- Any T/Any N M1c

Complications

Surgery Complications

- Air leak
- Pain
- Chest deformity
- Respiratory failure and/or death
- Injury to the phrenic nerve

Chemotherapy complications

- Pancytopenia
- Infections
- Hyponatremia
- Renal failure

- Peripheral neuropathy

Postoperative and Rehabilitation Care

Patients undergoing thoracotomy will need aggressive respiratory care including incentive spirometry, physical therapy, and chest therapy.

Consultations

Pulmonologist

Oncologist

Thoracic Surgeon

Intensivist

Cardiologist - for preoperative workup

Deterrence and Patient Education

- Smoking prevention
- Tobacco control
- Worker safety and use of proper equipment to prevent inhalation of toxic aerosols

Pearls and Other Issues

- The treatment of lung adenocarcinoma depends on the stage. For early-stage disease, surgery is the treatment of choice.
- For advanced disease, a combination of surgery, chemotherapy, and radiation is used to manage pain and other complications.
- Overall, outcomes for localized disease are far superior to other forms of treatment.
- Airway obstruction can be managed with laser and stent placement.
- Chemotherapy with targeted therapy may prolong survival, but the cost of the medications is prohibitive.
- The majority of lung cancer patients require palliation.
- Radiation therapy is often used to manage bone and brain metastases.

Enhancing Healthcare Team Outcomes

The majority of lung cancers are advanced at the time of diagnosis, and consequently, the prognosis is very poor. More than 80% of patients with advanced lung cancer are dead by 5 years. Despite all the advances, longevity has not increased over the past 3 decades. Thus, today the emphasis is on screening for lung cancer and prevention. Only an interprofessional team approach can help reduce the number of lung cancers and morbidity. Nurses and pharmacists are in a prime position to educate the public on smoking prevention. There are several antismoking aids that can be recommended by the pharmacist. Smoking cessation has repercussions beyond just reducing lung cancer; it can lower heart disease, stroke, and peripheral vascular disease.

Further, the public should be educated about work related to particle exposure (eg asbestos) and take appropriate precautions. Screening for lung cancer still remains a debatable topic. It is not recommended for everyone; as of today CT scan of the lung can be used to screen individuals over the age of 50 who have many risk factors for lung cancer. Whether the screening will help still remains to be seen. But it will definitely increase the cost of healthcare.[14][15] (level III)

Evidence-Based Outcomes (level V)

- Lung cancer is deadly with overall 5-year survival of 12%.[16][17][18]
- For Stage 1 disease, 5-year survival varies from 70% to 85%.
- For locally advanced disease, the survival drops to less than 30%.
- For distant metastases, less than 5% survive five years.

Review Questions

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